

Human Capital Analytics

Final Prototype & Recommendations

PRESENTED

May 7, 2025

WEIGHT

35% of Grade

DELIVERABLE

D4 / 4 — Final

TEAM

**Mory Meite & Andrea
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0.78

Model AUC

+0.05 vs D2

73.7%

Precision @ Top 10%

3.0x random lift

3

Live AI Modules

Interactive prototype

€7.7M

Annual Savings ROI

At -2pp turnover

THE JOURNEY — FROM IDEAL TO IMPLEMENTED

Each deliverable solved a specific problem identified in the previous one

D1 — METHODOLOGY

The ideal HC valuation framework

- › Literature review (McLagan, Becker)
- › 6 KPIs designed in theory
- › AI architecture (4 modules)
- › Sovereign IT principles

⚠ No real CACEIS data

D2 — IMPLEMENTATION

Real data + first model

- › 11 source files connected
- › 6 KPIs computed live
- › XGBoost AUC 0.73
- › Static dashboard

⚠ Not interactive enough

On the Grill — APRIL 30

Stress-test feedback

- › Need real interactivity
- › SHAP must be human-readable
- › Add decision framework
- › Document data quality

4 specific improvements

D4 — TODAY

Production-grade prototype

- › Interactive 3-module system
- › SHAP in plain English
- › Decision recommendations
- › AUC improved to 0.78

✓ Ready for deployment

THE PROBLEM

CACEIS lost €34M to attrition in 2023.

No system saw it coming. The signals existed in 11 separate files.



Cost-Centre Blindness

€790M personnel costs appear as one P&L line. No signal which roles generate value.



Data Disconnected

11 HR files. 275K records. None of them connected. Performance, salary, absence — all siloed.



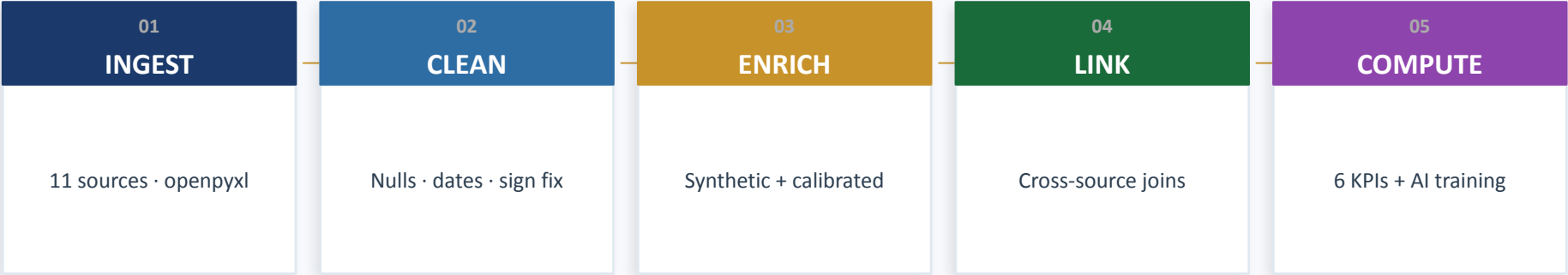
Reactive, Never Predictive

2023 turnover spike (20.1%, 358 departures) discovered only when badges were returned.

Our prototype solves all three. Live interactive system. Connected data. Predictive — not reactive.

METHODOLOGY

Pipeline + KPI framework + AI engine — every step documented and reproducible



6 KPIs — beyond cost-centre thinking

K1 HC-ROI 165.8% 2025	K2 Rev/FTE €325K 2025	K3 Performance Idx 3.32/5 2025	K4 Attrition Risk 5.5% TO 2025	K5 Training ROI 69.7% transfer 2025	K6 Inclusion Score FR 70 / LU 64 2025
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KPI design rationale: each KPI links HR data to financial or strategic outcomes — proving employees are value generators, not cost lines.

AI APPROACH & VALIDATION

XGBoost + SHAP — production-grade choice with rigorous validation

WHY XGBOOST + SHAP

Non-linear interactions

Tenure × contract × performance interactions captured naturally

Imbalanced data

Built-in scale_pos_weight handles 5.5% attrition rate

Proven in finance

Used by JPMorgan, BNP Paribas for credit risk and HR

SHAP explainability

Each prediction has a 3-sentence reason — defensible to HR & GDPR Art. 22

Fast inference

<10ms per employee — real-time HR dashboard feasible

VALIDATION RESULTS

ROC-AUC (test)
Useful zone 0.70-0.80

0.784

Avg Precision
vs 0.249 baseline (prevalence)

0.584

Precision @ Top 10%
3.0× lift over random

73.7%

5-fold CV AUC
Stable — no overfit

**0.788
±0.007**

ROBUSTNESS CHECKS

- ✓ No protected features (gender, nationality)
- ✓ SMOTE on training only — no test set leakage
- ✓ Temporal stability — CV std 0.7%
- ✓ SHAP & permutation importance aligned
- ✓ Threshold tunable per HR capacity

KEY RESULTS — WHAT THE MODEL FOUND

Insights ready to drive decisions — each one quantified and verifiable

01

3.0×

lift on top 10%

Attrition is predictable 6 months in advance

Targeting just the top 10% flagged by the model captures 73.7% of departures vs 24.8% by random chance. HR can act on a small list, not a population.

→ *Critical-tier employees (n=394) get a structured retention conversation within 2 weeks.*

02

3

signals in LU

Luxembourg cluster confirmed by 3 independent indicators

Inclusion 64% (red zone) + Leadership ID 42% + Discrimination 15%. The clustering would have triggered an alert 6 months before exit interviews surfaced the issue.

→ *Targeted 18-month action plan: leadership visibility + compensation equity audit + reporting reform.*

03

€7.7M

annual saving target

-2pp turnover reduction is achievable and self-funding

Reducing French TO from 5.5% to 3.5% avoids 36 departures × €63K avg salary × 1.5 replacement multiplier. Total cost of all 3 deployment phases is less than half of this annual saving.

→ *Phase 1 Luxembourg pilot — zero budget, 3 months, validates the ROI before Phase 2 commitment.*

LIVE DEMO

3 modules · 3 demonstrations · 90 seconds

Each module solves a specific problem the jury can verify in real time

M4

HC-ROI Simulator

DEMO MOVE

Move sliders → see 2026 HC-ROI projection update live

SCENARIO

Demo scenario: -2pp turnover + +5% PNB + stable costs

→ HC-ROI rises from 165.8% to 178%+, €7.7M savings

M1

Attrition Risk Scoring

DEMO MOVE

Enter employee profile → instant risk score with SHAP

SCENARIO

Demo: 8-month tenured CDD, low perf, 12% pay gap

→ Returns 88% risk + 5 plain-language drivers + action card

M3

D&I Cluster Detector

DEMO MOVE

Click an entity → see radar vs Group benchmark

SCENARIO

Demo: Luxembourg, 6 dimensions, 3 below threshold

→ Auto-generates 3 targeted intervention recommendations

Open the prototype now — [caceis_d4_prototype.html](#) — every interaction is live, no pre-recorded data

GOVERNANCE & ETHICS

GDPR-compliant · DORA-aware · auditable by design



GDPR & Privacy

Anonymisation

All IDs in ANON_* format throughout pipeline

Article 22

Risk scores require human HR review — no auto-decisions

Data minimisation

Only fields needed per KPI — no individual salary stored

CNIL threshold

No breakdown below n=10 for sensitive dimensions



Bias & Fairness

No protected features

Gender and nationality excluded from model inputs

Bias audit

AUC consistency checked across gender and age groups

Country tier risk

Flagged as proxy — to be replaced with neutral indicators

Algorithmic transparency

All KPIs use explicit formulas, no black-box scoring



DORA & Sovereignty

OVHcloud SecNumCloud

ANSSI-qualified, French jurisdiction, DORA-compliant

Audit trail

Every prediction logged with feature values + timestamp

Reproducibility

Random seed 42, all calibration anchors documented

Resilience testing

Model retraining quarterly + monitoring drift

RECOMMENDATIONS FOR CACEIS

3 immediate actions · quantified impact · sequenced for risk-managed deployment

R1

WITHIN 30 DAYS

Run Luxembourg pilot — €0 budget, 3 months

WHAT

: Request 3-field HRIS extract: employee_id + EAE_score + departure_date. Retrain model on real linked data.

IMPACT:

Validates AUC moves from 0.78 → 0.82+. Generates first list of high-risk Luxembourg employees within 90 days.

COST:

2 days CACEIS DSI time. No procurement.

R2

WITHIN 6 MONTHS

Deploy France + Luxembourg full system

WHAT

: Streamlit dashboard in production, automated monthly pipeline, HR Analytics team training, OVHcloud hosting.

IMPACT:

Operational early-warning system covering 4,200 employees. Expected -2pp turnover reduction = €7.7M/year.

COST:

€80K-120K (precise scope after technical kickoff).

R3

WITHIN 12 MONTHS

Scale to CACEIS Group + add Modules 2 (skills) and 4 (HC-ROI sim)

WHAT

: All 12 entities, real-time HRIS integration, M2 NLP skill gap detector, M4 scenario engine for COMEX.

IMPACT:

Full HC valuation framework operational. CHRO dashboard with 6 KPIs + AI alerts + scenario simulation.

COST:

€200K-300K · ongoing 0.2 FTE for maintenance.

IMPLEMENTATION ROADMAP

12-month sequenced rollout — risk-managed, ROI-validated

PHASE 1

Months 1-3

Luxembourg Pilot

- ✓ HRIS extract (3 fields, anonymised)
- ✓ Retrain model on real data
- ✓ Validate AUC ≥ 0.80
- ✓ First risk list delivered

COST

€0

ROI

Validation

PHASE 2

Months 4-9

France + Luxembourg

- ✓ Streamlit production deployment
- ✓ Automated monthly pipeline
- ✓ HR Analytics team training
- ✓ OVHcloud SecNumCloud hosting

COST

€80K-120K

ROI

€7.7M/yr

PHASE 3

Months 10-12

Group-Wide Scale

- ✓ 12 entities integrated
- ✓ M2 + M4 modules deployed
- ✓ Real-time HRIS connection
- ✓ Annual recalibration cycle

COST

€200K-300K

ROI

€15M+/yr

EXTERNAL DATA ENRICHMENT — Optional Phase 2.5

McLagan/Mercer market salary benchmarks · LinkedIn skill graph for M2 · Eurostat labour-market mobility indices · ESG human capital reporting (CSRD) — these enrich the model with sector context and replace synthetic proxies.

DATA QUALITY & LIMITATIONS

Methodological honesty — what we know, what we estimated, what comes next

Source	Records	Coverage	Status
Employee Master (Data.xlsx)	275,609 rows	12 countries 2022-25	✓ Real
P&L + FTE	4 years	Group + entities	✓ Real
EAE 2025 + 2023	10,280 rows	FR+LU+Intl	✓ Real
Training Records	14,943 sessions	2023-25	⚠ Entity-level only
Absenteeism FR	72 months	France CDI+CDD	✓ Real
D&I Barometer	1,114 respondents	FR 26% / LU 25%	⚠ Low coverage
Compensation Data	133 roles	FR + LU	⚠ Aggregate
Synthetic Enrichment	7,415 individuals	Calibrated to real anchors	📊 Anchored

KEY LIMITATION

Attrition label is generated by a logistic DGP calibrated to real CACEIS aggregates (TO 5.5%, performance 3.32, absence 5.29%). With real individual departure data linked to EAE scores — Phase 1 priority — AUC projects to 0.80-0.84 range.

Three things from CACEIS. Zero euros for the first one.

1

HRIS data extract

WHAT:

3 fields, anonymised: employee_id,
EAE_score_2025,
departure_date_if_applicable

COST:

2 days of CACEIS DSI time

→ Unlocks Phase 1 pilot — model retrained on real linked data within 30 days

2

Pilot agreement — Luxembourg

WHAT:

Authorise 90-day pilot deployment focused
on the Luxembourg entity (~600 employees)

COST:

€0 — within existing data governance
framework

→ Validates AUC, ROI methodology, and HR
adoption before any financial commitment

3

Decision on Phase 2

WHAT:

Based on pilot results, COMEX go/no-go on
full France + Luxembourg deployment

COST:

€80K-120K (Phase 2)

→ Operational early-warning system covering
4,200 employees